```
> # PROOF OF IDENTITIES FOR a_4(n)
  with(qseries):
  with(ETA):
  t := gp2etaprod([7, 4, 1, -4]):
  p0 := gp2etaprod([14, 4, 1, 4, 7, -4, 2, -4]):
  p1 := (gp2etaprod([14, 1, 1, 7, 7, -1, 2, -7]) - 1)/7:
  A4 := gp2etaprod([2, 3, 49, 4, 1, -4, 98, -3]):
  B4 := 1:

> # GROUP 1, IDENTITY 1: U_A4(p0) = 7^2*t^2*p0 + 7*t
  EP4_1_1 := normal(A4*p0):
  RHS4_1_1 := expand(7^2*t^2*p0 + 7*t):
  proof4_1_1 := provemodfuncGAMMA0UpETAidBATCH(EP4_1_1, 7, RHS4_1_1, 14);

*** There were NO errors.
*** o each EP in gcombo is an MF on Gamma[0](98)
*** o Each term in the etacombo is a  modular function on
      Gamma0(14).
*** o We also checked that the total order of
      each term etacombo was zero.
*** To prove the identity U[7](EP)=etacombo we need to show
    that v[oo](ID) > 4    This means checking up to q^(5).
We find that LHS - RHS is

                                  O(q^32)
proof4_1_1 := [1, -4, O(q^32)]                                  (1)

> # GROUP 1, IDENTITY 2: U_A4(p0*t^(-1))
  EP4_1_2 := normal(A4*p0/t):
  RHS4_1_2 := expand(-7*t*p0 + p1 - 7*t - 1):
  proof4_1_2 := provemodfuncGAMMA0UpETAidBATCH(EP4_1_2, 7, RHS4_1_2, 14);

*** There were NO errors.
*** o each EP in gcombo is an MF on Gamma[0](98)
*** o Each term in the etacombo is a  modular function on
      Gamma0(14).
*** o We also checked that the total order of
      each term etacombo was zero.
*** To prove the identity U[7](EP)=etacombo we need to show
    that v[oo](ID) > 5    This means checking up to q^(6).
We find that LHS - RHS is

                                  O(q^33)
proof4_1_2 := [1, -5, O(q^33)]                                  (2)
```

```
> # GROUP 1, IDENTITY 3: U_A4(p0*t^(-2))
  EP4_1_3 := normal(A4*p0/t^2):
  RHS4_1_3 := expand(p0*(7^5*t^3+2*7^4*t^2+54*7*t)+p1*(-7^4*t^2-13*7
      ^2*t-48)+7^4*t^2+95*7*t+48):
  proof4_1_3 := provemodfuncGAMMA0UpETAidBATCH(EP4_1_3, 7, RHS4_1_3, 14);

*** There were NO errors.
*** o each EP in gcombo is an MF on Gamma[0](98)
*** o Each term in the etacombo is a  modular function on
      Gamma0(14).
*** o We also checked that the total order of
      each term etacombo was zero.
*** To prove the identity U[7](EP)=etacombo we need to show
    that v[oo](ID) > 9    This means checking up to q^(10).
We find that LHS - RHS is
```

$$\mathrm{O}(q^{37})$$

$$proof4_1_3 := [1, -9, \mathrm{O}(q^{37})] \quad \textbf{(3)}$$

```
> # GROUP 1, IDENTITY 4: U_A4(p0*t^(-3))
  EP4_1_4 := normal(A4*p0/t^3):
  RHS4_1_4 := expand(p0*(7^7*t^4+2*7^5*t^3-13*7^4*t^2-459*7*t)+p1*(-7
      ^6*t^3-7^4*t^2+87*7^2*t+58*7)+5*7^5*t^3-3*7^3*t^2-90*7^2*t-58
      *7):
  proof4_1_4 := provemodfuncGAMMA0UpETAidBATCH(EP4_1_4, 7, RHS4_1_4, 14);

*** There were NO errors.
*** o each EP in gcombo is an MF on Gamma[0](98)
*** o Each term in the etacombo is a  modular function on
      Gamma0(14).
*** o We also checked that the total order of
      each term etacombo was zero.
*** To prove the identity U[7](EP)=etacombo we need to show
    that v[oo](ID) > 12    This means checking up to q^(13).
We find that LHS - RHS is
```

$$\mathrm{O}(q^{40})$$

$$proof4_1_4 := [1, -12, \mathrm{O}(q^{40})] \quad \textbf{(4)}$$

```
> # GROUP 1, IDENTITY 5: U_A4(p0*t^(-4))
  EP4_1_5 := normal(A4*p0/t^4):
  RHS4_1_5 := expand(p0*(-7^9*t^5-36*7^7*t^4-38*7^6*t^3-181*7^3*t^2+363
      *7^2*t+8)+p1*(7^8*t^4+5*7^7*t^3+34*7^5*t^2+16*7^3*t-321*7-t
      ^(-1))-7^8*t^4-31*7^6*t^3-218*7^4*t^2-95*7^2*t+321*7+t^(-1)):
```

```
proof4_1_5 := provemodfuncGAMMA0UpETAidBATCH(EP4_1_5, 7, RHS4_1_5, 14);
```

```
*** There were NO errors.
*** o each EP in gcombo is an MF on Gamma[0](98)
*** o Each term in the etacombo is a  modular function on
      Gamma0(14).
*** o We also checked that the total order of
      each term etacombo was zero.
*** To prove the identity U[7](EP)=etacombo we need to show
    that v[oo](ID) > 17    This means checking up to q^(18).
We find that LHS - RHS is
```

$$O(q^{45})$$

$$proof4_1_5 := [1, -17, O(q^{45})] \quad \textbf{(5)}$$

```
> # GROUP 1, IDENTITY 6: U_A4(p0*t^(-5))
  EP4_1_6 := normal(A4 * p0 / t^5) :
  RHS4_1_6 := expand(p0 * (7^11 * t^6 + 41 * 7^9 * t^5 + 734 * 7^7 * t^4 + 727 * 7^6 * t^3
      + 1602 * 7^4 * t^2 - 1025 * 7^2 * t - 160) + p1 * (-7^10 * t^5 - 40 * 7^8 * t^4 - 100 * 7^7
      * t^3 - 659 * 7^5 * t^2 - 1374 * 7^3 * t + 913 * 7 + 20 * t^(-1)) + 6 * 7^9 * t^5 + 265 * 7
      ^7 * t^4 + 93 * 7^7 * t^3 + 4404 * 7^4 * t^2 + 9519 * 7^2 * t - 6390 - 20 * t^(-1)) :
  proof4_1_6 := provemodfuncGAMMA0UpETAidBATCH(EP4_1_6, 7, RHS4_1_6, 14);
```

```
*** There were NO errors.
*** o each EP in gcombo is an MF on Gamma[0](98)
*** o Each term in the etacombo is a  modular function on
      Gamma0(14).
*** o We also checked that the total order of
      each term etacombo was zero.
*** To prove the identity U[7](EP)=etacombo we need to show
    that v[oo](ID) > 20    This means checking up to q^(21).
We find that LHS - RHS is
```

$$O(q^{48})$$

$$proof4_1_6 := [1, -20, O(q^{48})] \quad \textbf{(6)}$$

```
> # GROUP 1, IDENTITY 7: U_A4(p0*t^(-6))
  EP4_1_7 := normal(A4 * p0 / t^6) :
  RHS4_1_7 := expand(p0 * (-7^13 * t^7 - 46 * 7^11 * t^6 - 947 * 7^9 * t^5 - 11264 * 7^7 * t^4
      - 9647 * 7^6 * t^3 - 23659 * 7^4 * t^2 - 6634 * 7^2 * t + 2024) + p1 * (7^12 * t^6 + 45 * 7
      ^10 * t^5 + 907 * 7^8 * t^4 + 216 * 7^8 * t^3 + 8747 * 7^5 * t^2 + 20686 * 7^3 * t + 5744
      * 7 - 253 * t^(-1)) - 5 * 7^11 * t^6 - 258 * 7^9 * t^5 - 828 * 7^8 * t^4 - 1423 * 7^7 * t^3
      - 59252 * 7^4 * t^2 - 144218 * 7^2 * t - 821 * 7^2 + 253 * t^(-1)) :
  proof4_1_7 := provemodfuncGAMMA0UpETAidBATCH(EP4_1_7, 7, RHS4_1_7, 14);
```

```
*** There were NO errors.
*** o each EP in gcombo is an MF on Gamma[0](98)
*** o Each term in the etacombo is a  modular function on
      Gamma0(14).
*** o We also checked that the total order of
      each term etacombo was zero.
*** To prove the identity U[7](EP)=etacombo we need to show
    that v[oo](ID) > 23    This means checking up to q^(24).
We find that LHS - RHS is
```

$$\mathrm{O}(q^{51})$$

$$proof4_1_7 := [1, -23, \mathrm{O}(q^{51})] \quad \textbf{(7)}$$

```
> # GROUP 2, IDENTITY 1: U_A4(p1)
  EP4_2_1 := expand(A4*p1):
  RHS4_2_1 := expand(p0*(8*7^11*t^7+144*7^9*t^6+139*7^8*t^5+404*7^6*t^4
      +417*7^4*t^3-15*7^2*t^2-8*7*t)+p1*(7^14*t^8+27*7^12*t^7+293*7
      ^10*t^6+1604*7^8*t^5+646*7^7*t^4+845*7^5*t^3+8*7^5*t^2+114*7
      *t+7)+7^13*t^8+19*7^11*t^7+141*7^9*t^6+69*7^8*t^5+99*7^6*t^4
      -12*7^4*t^3-297*7^2*t^2-104*7*t-7):
  proof4_2_1 := provemodfuncGAMMA0UpETAidBATCH(EP4_2_1,7,RHS4_2_1,14);
```

```
*** There were NO errors.
*** o each EP in gcombo is an MF on Gamma[0](98)
*** o Each term in the etacombo is a  modular function on
      Gamma0(14).
*** o We also checked that the total order of
      each term etacombo was zero.
*** To prove the identity U[7](EP)=etacombo we need to show
    that v[oo](ID) > 24    This means checking up to q^(25).
We find that LHS - RHS is
```

$$\mathrm{O}(q^{52})$$

$$proof4_2_1 := [1, -24, \mathrm{O}(q^{52})] \quad \textbf{(8)}$$

```
> # GROUP 2, IDENTITY 2: U_A4(p1*t^(-1))
  EP4_2_2 := expand(A4*p1/t):
  RHS4_2_2 := expand(-8*7*t*p0+(7^2*t+7)*p1-7):
  proof4_2_2 := provemodfuncGAMMA0UpETAidBATCH(EP4_2_2,7,RHS4_2_2,14);
```

```
*** There were NO errors.
*** o each EP in gcombo is an MF on Gamma[0](98)
*** o Each term in the etacombo is a  modular function on
      Gamma0(14).
```

```
*** o We also checked that the total order of
      each term etacombo was zero.
*** To prove the identity U[7](EP)=etacombo we need to show
    that v[oo](ID) > 4    This means checking up to q^(5).
We find that LHS - RHS is
```

$$\mathrm{O}(q^{32})$$

$$proof4_2_2 := [1, -4, \mathrm{O}(q^{32})] \quad \textbf{(9)}$$

```
> # GROUP 2, IDENTITY 3: U_A4(p1*t^(-2))
  EP4_2_3 := expand(A4*p1/t^2):
  RHS4_2_3 := expand(p0*(3*7^3*t^2+11*7*t)+p1*(-3*7^2*t-9)+7^3*t^2
      +18*7*t+9):
  proof4_2_3 := provemodfuncGAMMA0UpETAidBATCH(EP4_2_3,7,RHS4_2_3,14);
```

```
*** There were NO errors.
*** o each EP in gcombo is an MF on Gamma[0](98)
*** o Each term in the etacombo is a  modular function on
      Gamma0(14).
*** o We also checked that the total order of
      each term etacombo was zero.
*** To prove the identity U[7](EP)=etacombo we need to show
    that v[oo](ID) > 7    This means checking up to q^(8).
We find that LHS - RHS is
```

$$\mathrm{O}(q^{35})$$

$$proof4_2_3 := [1, -7, \mathrm{O}(q^{35})] \quad \textbf{(10)}$$

```
> # GROUP 2, IDENTITY 4: U_A4(p1*t^(-3))
  EP4_2_4 := expand(A4*p1/t^3):
  RHS4_2_4 := expand(p0*(2*7^5*t^3+27*7^3*t^2+131*7*t)+p1*(-2*7^4*t^2
      -24*7^2*t-118)-7^5*t^3+4*7^3*t^2+25*7^2*t+118):
  proof4_2_4 := provemodfuncGAMMA0UpETAidBATCH(EP4_2_4,7,RHS4_2_4,14);
```

```
*** There were NO errors.
*** o each EP in gcombo is an MF on Gamma[0](98)
*** o Each term in the etacombo is a  modular function on
      Gamma0(14).
*** o We also checked that the total order of
      each term etacombo was zero.
*** To prove the identity U[7](EP)=etacombo we need to show
    that v[oo](ID) > 10    This means checking up to q^(11).
We find that LHS - RHS is
```

$$O(q^{38})$$

$$proof4_2_4 := [1, -10, O(q^{38})] \quad \textbf{(11)}$$

```
> # GROUP 2, IDENTITY 5: U_A4(p1*t^(-4))
  EP4_2_5 := expand(A4*p1/t^4):
  RHS4_2_5 := expand(p0*(2*7^7*t^4-256*7^3*t^2-206*7^2*t-8)+p1*(-2*7^6
      *t^3+3*7^4*t^2+241*7^2*t+183*7+t^(-1))-7^7*t^4+4*7^5*t^3-23
      *7^3*t^2-247*7^2*t-183*7-t^(-1)):
  proof4_2_5 := provemodfuncGAMMA0UpETAidBATCH(EP4_2_5,7,RHS4_2_5,14);
```

```
*** There were NO errors.
*** o each EP in gcombo is an MF on Gamma[0](98)
*** o Each term in the etacombo is a  modular function on
      Gamma0(14).
*** o We also checked that the total order of
      each term etacombo was zero.
*** To prove the identity U[7](EP)=etacombo we need to show
    that v[oo](ID) > 15    This means checking up to q^(16).
We find that LHS - RHS is
```

$$O(q^{43})$$

$$proof4_2_5 := [1, -15, O(q^{43})] \quad \textbf{(12)}$$

```
> # GROUP 2, IDENTITY 6: U_A4(p1*t^(-5))
  EP4_2_6 := expand(A4*p1/t^5):
  RHS4_2_6 := expand(p0*(-5*7^9*t^5-138*7^7*t^4-143*7^6*t^3-171*7^4*t^2
      +1192*7^2*t+152)+p1*(5*7^8*t^4+19*7^7*t^3+127*7^5*t^2+130*7
      ^3*t-1058*7-19*t^(-1))-22*7^7*t^4-106*7^6*t^3-814*7^4*t^2-862*7
      ^2*t+7407+19*t^(-1)):
  proof4_2_6 := provemodfuncGAMMA0UpETAidBATCH(EP4_2_6,7,RHS4_2_6,14);
```

```
*** There were NO errors.
*** o each EP in gcombo is an MF on Gamma[0](98)
*** o Each term in the etacombo is a  modular function on
      Gamma0(14).
*** o We also checked that the total order of
      each term etacombo was zero.
*** To prove the identity U[7](EP)=etacombo we need to show
    that v[oo](ID) > 18    This means checking up to q^(19).
We find that LHS - RHS is
```

$$O(q^{46})$$

$$proof4_2_6 := [1, -18, O(q^{46})] \quad \textbf{(13)}$$

```
> # GROUP 2, IDENTITY 7: U_A4(p1*t^(-6))
```

```
EP4_2_7 := expand(A4*p1/t^6):
RHS4_2_7 := expand(p0*(8*7^11*t^6+38*7^10*t^5+3726*7^7*t^4+3298*7^6*t
    ^3+7423*7^4*t^2-2693*7^2*t-1864)+p1*(-8*7^10*t^5-37*7^9*t^4-72
    *7^8*t^3-2978*7^5*t^2-6404*7^3*t+2431*7+233*t^(-1)))+7^11*t^6
    +68*7^9*t^5+1727*7^7*t^4+3238*7^6*t^3+19876*7^4*t^2+907*7^4*t
    -17029-233*t^(-1)):
proof4_2_7 := provemodfuncGAMMA0UpETAidBATCH(EP4_2_7,7,RHS4_2_7,14);
```

```
*** There were NO errors.
*** o each EP in gcombo is an MF on Gamma[0](98)
*** o Each term in the etacombo is a  modular function on
      Gamma0(14).
*** o We also checked that the total order of
      each term etacombo was zero.
*** To prove the identity U[7](EP)=etacombo we need to show
    that v[oo](ID) > 21    This means checking up to q^(22).
We find that LHS - RHS is
```

$$O(q^{49})$$

$$proof4_2_7 := [1, -21, O(q^{49})] \quad \textbf{(14)}$$

```
> # GROUP 3, IDENTITY 1: U_A4(1)
EP4_3_1 := normal(A4):
RHS4_3_1 := expand(p0*(3*7^8*t^5+52*7^6*t^4+321*7^4*t^3+111*7^3*t^2
    +72*7*t)+p1*(-3*7^7*t^4-50*7^5*t^3-297*7^3*t^2-99*7^2*t-9*7)
    +7^8*t^5+34*7^6*t^4+405*7^4*t^3+2150*7^2*t^2+695*7*t+9*7):
proof4_3_1 := provemodfuncGAMMA0UpETAidBATCH(EP4_3_1,7,RHS4_3_1,14);
```

```
*** There were NO errors.
*** o each EP in gcombo is an MF on Gamma[0](98)
*** o Each term in the etacombo is a  modular function on
      Gamma0(14).
*** o We also checked that the total order of
      each term etacombo was zero.
*** To prove the identity U[7](EP)=etacombo we need to show
    that v[oo](ID) > 16    This means checking up to q^(17).
We find that LHS - RHS is
```

$$O(q^{44})$$

$$proof4_3_1 := [1, -16, O(q^{44})] \quad \textbf{(15)}$$

```
> # GROUP 3, IDENTITY 2: U_A4(t^(-1))
EP4_3_2 := normal(A4/t):
RHS4_3_2 := -7*t:
```

```
proof4_3_2 := provemodfuncGAMMA0UpETAidBATCH(EP4_3_2, 7, RHS4_3_2, 14);
```

```
*** There were NO errors.
*** o each EP in gcombo is an MF on Gamma[0](98)
*** o Each term in the etacombo is a  modular function on
      Gamma0(14).
*** o We also checked that the total order of
      each term etacombo was zero.
*** To prove the identity U[7](EP)=etacombo we need to show
    that v[oo](ID) > 1    This means checking up to q^(2).
We find that LHS - RHS is
```

$$O(q^{29})$$

$$proof4_3_2 := [1, -1, O(q^{29})] \quad \textbf{(16)}$$

```
> # GROUP 3, IDENTITY 3: U_A4(t^(-2))
  EP4_3_3 := normal(A4/t^2):
  RHS4_3_3 := expand(p0*(3*7^3*t^2+11*7*t)+p1*(-3*7^2*t-10)+7^3*t^2
      +26*7*t+10):
  proof4_3_3 := provemodfuncGAMMA0UpETAidBATCH(EP4_3_3, 7, RHS4_3_3, 14);
```

```
*** There were NO errors.
*** o each EP in gcombo is an MF on Gamma[0](98)
*** o Each term in the etacombo is a  modular function on
      Gamma0(14).
*** o We also checked that the total order of
      each term etacombo was zero.
*** To prove the identity U[7](EP)=etacombo we need to show
    that v[oo](ID) > 7    This means checking up to q^(8).
We find that LHS - RHS is
```

$$O(q^{35})$$

$$proof4_3_3 := [1, -7, O(q^{35})] \quad \textbf{(17)}$$

```
> # GROUP 3, IDENTITY 4: U_A4(t^(-3))
  EP4_3_4 := normal(A4/t^3):
  RHS4_3_4 := expand(p0*(-6*7^5*t^3-85*7^3*t^2-5*7^3*t+8)+p1*(6*7^4*t
      ^2+80*7^2*t+31*7-t^(-1))-7^5*t^3-52*7^3*t^2-585*7*t-31*7+t^(
      -1)):
  proof4_3_4 := provemodfuncGAMMA0UpETAidBATCH(EP4_3_4, 7, RHS4_3_4, 14);
```

```
*** There were NO errors.
*** o each EP in gcombo is an MF on Gamma[0](98)
*** o Each term in the etacombo is a  modular function on
      Gamma0(14).
```

```
*** o We also checked that the total order of
      each term etacombo was zero.
*** To prove the identity U[7](EP)=etacombo we need to show
    that v[oo](ID) > 12    This means checking up to q^(13).
We find that LHS - RHS is
```

$$O(q^{40})$$

$$proof4_3_4 := [1, -12, O(q^{40})] \quad \textbf{(18)}$$

```
> # GROUP 3, IDENTITY 5: U_A4(t^(-4))
  EP4_3_5 := normal(A4/t^4) :
  RHS4_3_5 := expand(p0*(-6*7^7*t^4-8*7^5*t^3+80*7^4*t^2+306*7^2*t-96)
      +p1*(6*7^6*t^3+4*7^4*t^2-76*7^3*t-270*7+12*t^(-1)))-7^7*t^4
      -36*7^5*t^3+50*7^3*t^2+550*7^2*t+270*7-12*t^(-1)) :
  proof4_3_5 := provemodfuncGAMMA0UpETAidBATCH(EP4_3_5, 7, RHS4_3_5, 14);
```

```
*** There were NO errors.
*** o each EP in gcombo is an MF on Gamma[0](98)
*** o Each term in the etacombo is a  modular function on
      Gamma0(14).
*** o We also checked that the total order of
      each term etacombo was zero.
*** To prove the identity U[7](EP)=etacombo we need to show
    that v[oo](ID) > 15    This means checking up to q^(16).
We find that LHS - RHS is
```

$$O(q^{43})$$

$$proof4_3_5 := [1, -15, O(q^{43})] \quad \textbf{(19)}$$

```
> # GROUP 3, IDENTITY 6: U_A4(t^(-5))
  EP4_3_6 := normal(A4/t^5) :
  RHS4_3_6 := expand(p0*(3*7^9*t^5+158*7^7*t^4+23*7^7*t^3-67*7^4*t^2
      -1872*7^2*t+808)+p1*(-3*7^8*t^4-22*7^7*t^3-145*7^5*t^2+92*7^3
      *t+1650*7-101*t^(-1)))+34*7^7*t^4+149*7^6*t^3+930*7^4*t^2-732
      *7^2*t-11549+101*t^(-1))) :
  proof4_3_6 := provemodfuncGAMMA0UpETAidBATCH(EP4_3_6, 7, RHS4_3_6, 14);
```

```
*** There were NO errors.
*** o each EP in gcombo is an MF on Gamma[0](98)
*** o Each term in the etacombo is a  modular function on
      Gamma0(14).
*** o We also checked that the total order of
      each term etacombo was zero.
*** To prove the identity U[7](EP)=etacombo we need to show
```

```
    that v[oo](ID) > 18    This means checking up to q^(19).
We find that LHS - RHS is
```

$$O(q^{46})$$

$$proof4_3_6 := [1, -18, O(q^{46})] \quad \textbf{(20)}$$

```
> # GROUP 3, IDENTITY 7: U_A4(t^(-6))
  EP4_3_7 := normal(A4/t^6):
  RHS4_3_7 := expand(p0*(-10*7^10*t^5-50*7^9*t^4-2822*7^6*t^3-5545*7^4*t
      ^2+7179*7^2*t-800*7)+p1*(68*7^8*t^4+334*7^7*t^3+2560*7^5*t^2
      +4721*7^3*t-6336*7+100*7*t^(-1))+7^11*t^6+20*7^9*t^5-442*7^7
      *t^4-2236*7^6*t^3-17218*7^4*t^2-32640*7^2*t+44332-100*7*t^(-1)):
  proof4_3_7 := provemodfuncGAMMA0UpETAidBATCH(EP4_3_7,7,RHS4_3_7,14);
*** There were NO errors.
*** o each EP in gcombo is an MF on Gamma[0](98)
*** o Each term in the etacombo is a  modular function on
      Gamma0(14).
*** o We also checked that the total order of
      each term etacombo was zero.
*** To prove the identity U[7](EP)=etacombo we need to show
    that v[oo](ID) > 20    This means checking up to q^(21).
We find that LHS - RHS is
```

$$O(q^{48})$$

$$proof4_3_7 := [1, -20, O(q^{48})] \quad \textbf{(21)}$$

```
> # GROUP 4, IDENTITY 1: U_B4(p0)
  EP4_4_1 := normal(B4*p0):
  RHS4_4_1 := expand(p0*(8*7^7*t^4+80*7^5*t^3+216*7^3*t^2+79*7*t)+p1
      *(7^10*t^5+19*7^8*t^4+18*7^7*t^3+327*7^4*t^2+34*7^3*t+4)+7^9
      *t^5+11*7^7*t^4+38*7^5*t^3+31*7^3*t^2+6*7*t):
  proof4_4_1 := provemodfuncGAMMA0UpETAidBATCH(EP4_4_1,7,RHS4_4_1,14);
*** There were NO errors.
*** o each EP in gcombo is an MF on Gamma[0](98)
*** o Each term in the etacombo is a  modular function on
      Gamma0(14).
*** o We also checked that the total order of
      each term etacombo was zero.
*** To prove the identity U[7](EP)=etacombo we need to show
    that v[oo](ID) > 15    This means checking up to q^(16).
We find that LHS - RHS is
```

$$O(q^{43})$$

$$proof4_4_1 := [1, -15, O(q^{43})] \quad \textbf{(22)}$$

```
> # GROUP 4, IDENTITY 2: U_B4(p0*t^(-1))
  EP4_4_2 := normal(B4*p0/t):
  RHS4_4_2 := expand(p0*(-7^3*t^2-8*7*t)+p1*(7^2*t+7)+1-7^2*t):
  proof4_4_2 := provemodfuncGAMMA0UpETAidBATCH(EP4_4_2, 7, RHS4_4_2, 14);
```

```
*** There were NO errors.
*** o each EP in gcombo is an MF on Gamma[0](98)
*** o Each term in the etacombo is a  modular function on
      Gamma0(14).
*** o We also checked that the total order of
      each term etacombo was zero.
*** To prove the identity U[7](EP)=etacombo we need to show
    that v[oo](ID) > 6    This means checking up to q^(7).
We find that LHS - RHS is
```

$$O(q^{34})$$

$$proof4_4_2 := [1, -6, O(q^{34})] \quad \textbf{(23)}$$

```
> # GROUP 4, IDENTITY 3: U_B4(p0*t^(-2))
  EP4_4_3 := normal(B4*p0/t^2):
  RHS4_4_3 := expand(p0*(-7^5*t^3-8*7^3*t^2)+p1*(7^4*t^2+7^3*t)-7^4*t^2
      -7^3*t-12):
  proof4_4_3 := provemodfuncGAMMA0UpETAidBATCH(EP4_4_3, 7, RHS4_4_3, 14);
```

```
*** There were NO errors.
*** o each EP in gcombo is an MF on Gamma[0](98)
*** o Each term in the etacombo is a  modular function on
      Gamma0(14).
*** o We also checked that the total order of
      each term etacombo was zero.
*** To prove the identity U[7](EP)=etacombo we need to show
    that v[oo](ID) > 9    This means checking up to q^(10).
We find that LHS - RHS is
```

$$O(q^{37})$$

$$proof4_4_3 := [1, -9, O(q^{37})] \quad \textbf{(24)}$$

```
> # GROUP 4, IDENTITY 4: U_B4(p0*t^(-3))
  EP4_4_4 := normal(B4*p0/t^3):
  RHS4_4_4 := expand(p0*(1+6*7^7*t^4+62*7^5*t^3+27*7^4*t^2+10*7^2*t)
      +p1*(-6*7^6*t^3-8*7^5*t^2-3*7^4*t-4*7)+2*7^6*t^3+24*7^4*t^2
      +83*7^2*t+108):
```

```
proof4_4_4 := provemodfuncGAMMA0UpETAidBATCH(EP4_4_4, 7, RHS4_4_4, 14);
```

```
*** There were NO errors.
*** o each EP in gcombo is an MF on Gamma[0](98)
*** o Each term in the etacombo is a  modular function on
      Gamma0(14).
*** o We also checked that the total order of
      each term etacombo was zero.
*** To prove the identity U[7](EP)=etacombo we need to show
    that v[oo](ID) > 13    This means checking up to q^(14).
We find that LHS - RHS is
```

$$O(q^{41})$$

$$proof4_4_4 := [1, -13, O(q^{41})] \quad \textbf{(25)}$$

```
> # GROUP 4, IDENTITY 5: U_B4(p0*t^(-4))
  EP4_4_5 := normal(B4*p0/t^4):
  RHS4_4_5 := expand(p0*(-7^9*t^5-50*7^7*t^4-60*7^6*t^3-162*7^4*t^2-60
      *7^2*t-6)+p1*(7^8*t^4+7^8*t^3+54*7^5*t^2+18*7^4*t+24*7)-7^8
      *t^4-25*7^6*t^3-186*7^4*t^2-498*7^2*t-120*7):
  proof4_4_5 := provemodfuncGAMMA0UpETAidBATCH(EP4_4_5, 7, RHS4_4_5, 14);
```

```
*** There were NO errors.
*** o each EP in gcombo is an MF on Gamma[0](98)
*** o Each term in the etacombo is a  modular function on
      Gamma0(14).
*** o We also checked that the total order of
      each term etacombo was zero.
*** To prove the identity U[7](EP)=etacombo we need to show
    that v[oo](ID) > 16    This means checking up to q^(17).
We find that LHS - RHS is
```

$$O(q^{44})$$

$$proof4_4_5 := [1, -16, O(q^{44})] \quad \textbf{(26)}$$

```
> # GROUP 4, IDENTITY 6: U_B4(p0*t^(-5))
  EP4_4_6 := normal(B4*p0/t^5):
  RHS4_4_6 := expand(p0*(-7^11*t^6-8*7^9*t^5+3*7^9*t^4+30*7^7*t^3+81
      *7^5*t^2+30*7^3*t+3*7)+p1*(7^10*t^5+7^9*t^4-3*7^8*t^3-27*7^6
      *t^2-9*7^5*t-12*7^2)-7^10*t^5-7^9*t^4+9*7^7*t^3+93*7^5*t^2
      +249*7^3*t+836*7):
  proof4_4_6 := provemodfuncGAMMA0UpETAidBATCH(EP4_4_6, 7, RHS4_4_6, 14);
```

```
*** There were NO errors.
*** o each EP in gcombo is an MF on Gamma[0](98)
```

```
*** o Each term in the etacombo is a  modular function on
      Gamma0(14).
*** o We also checked that the total order of
      each term etacombo was zero.
*** To prove the identity U[7](EP)=etacombo we need to show
    that v[oo](ID) > 19    This means checking up to q^(20).
We find that LHS - RHS is
```

$$O(q^{47})$$

$$proof4_4_6 := [1, -19, O(q^{47})] \quad \textbf{(27)}$$

```
> # GROUP 4, IDENTITY 7: U_B4(p0*t^(-6))
  EP4_4_7 := normal(B4*p0/t^6):
  RHS4_4_7 := expand(p0*(-7^13*t^7-8*7^11*t^6+2*7^9*t^4+20*7^7*t^3+54
      *7^5*t^2+20*7^3*t+2*7)+p1*(7^12*t^6+7^11*t^5-2*7^8*t^3-18*7
      ^6*t^2-6*7^5*t-8*7^2)-7^12*t^6-7^11*t^5+6*7^7*t^3+62*7^5*t^2
      +166*7^3*t-748*7^2):
  proof4_4_7 := provemodfuncGAMMA0UpETAidBATCH(EP4_4_7,7,RHS4_4_7,14);
```

```
*** There were NO errors.
*** o each EP in gcombo is an MF on Gamma[0](98)
*** o Each term in the etacombo is a  modular function on
      Gamma0(14).
*** o We also checked that the total order of
      each term etacombo was zero.
*** To prove the identity U[7](EP)=etacombo we need to show
    that v[oo](ID) > 22    This means checking up to q^(23).
We find that LHS - RHS is
```

$$O(q^{50})$$

$$proof4_4_7 := [1, -22, O(q^{50})] \quad \textbf{(28)}$$

```
> # GROUP 5, IDENTITY 1: U_B4(p1*t^(-1))
  EP4_5_1 := expand(B4*p1/t):
  RHS4_5_1 := expand(p0*(8*7^8*t^4+80*7^6*t^3+216*7^4*t^2+552*7*t)+p1
      *(7^11*t^5+19*7^9*t^4+18*7^8*t^3+327*7^5*t^2+34*7^4*t+29)-1
      +7^10*t^5+11*7^8*t^4+38*7^6*t^3+31*7^4*t^2+41*7*t):
  proof4_5_1 := provemodfuncGAMMA0UpETAidBATCH(EP4_5_1,7,RHS4_5_1,14);
```

```
*** There were NO errors.
*** o each EP in gcombo is an MF on Gamma[0](98)
*** o Each term in the etacombo is a  modular function on
      Gamma0(14).
*** o We also checked that the total order of
```

```
      each term etacombo was zero.
*** To prove the identity U[7](EP)=etacombo we need to show
    that v[oo](ID) > 15    This means checking up to q^(16).
We find that LHS - RHS is
```

$$O(q^{43})$$

$$proof4_5_1 := [1, -15, O(q^{43})] \quad \textbf{(29)}$$

```
> # GROUP 5, IDENTITY 2: U_B4(p1*t^(-2))
  EP4_5_2 := expand(B4*p1/t^2):
  RHS4_5_2 := expand(p0*(-8*7^3*t^2-8*7^2*t)+p1*(8*7^2*t+7^2)-7^3*t^2
      -8*7^2*t+11):
  proof4_5_2 := provemodfuncGAMMA0UpETAidBATCH(EP4_5_2,7,RHS4_5_2,14);
*** There were NO errors.
*** o each EP in gcombo is an MF on Gamma[0](98)
*** o Each term in the etacombo is a  modular function on
      Gamma0(14).
*** o We also checked that the total order of
      each term etacombo was zero.
*** To prove the identity U[7](EP)=etacombo we need to show
    that v[oo](ID) > 7    This means checking up to q^(8).
We find that LHS - RHS is
```

$$O(q^{35})$$

$$proof4_5_2 := [1, -7, O(q^{35})] \quad \textbf{(30)}$$

```
> # GROUP 5, IDENTITY 3: U_B4(p1*t^(-3))
  EP4_5_3 := expand(B4*p1/t^3):
  RHS4_5_3 := expand(p0*(-8*7^5*t^3-8*7^4*t^2)+p1*(8*7^4*t^2+7^4*t)-7
      ^5*t^3-8*7^4*t^2-7^4*t-96):
  proof4_5_3 := provemodfuncGAMMA0UpETAidBATCH(EP4_5_3,7,RHS4_5_3,14);
*** There were NO errors.
*** o each EP in gcombo is an MF on Gamma[0](98)
*** o Each term in the etacombo is a  modular function on
      Gamma0(14).
*** o We also checked that the total order of
      each term etacombo was zero.
*** To prove the identity U[7](EP)=etacombo we need to show
    that v[oo](ID) > 10    This means checking up to q^(11).
We find that LHS - RHS is
```

$$O(q^{38})$$

(31)

$$proof4_5_3 := [1, -10, O(q^{38})] \quad \textbf{(31)}$$

```
> # GROUP 5, IDENTITY 4: U_B4(p1*t^(-4))
  EP4_5_4 := expand(B4*p1/t^4):
  RHS4_5_4 := expand(p0*(48*7^7*t^4+72*7^6*t^3+216*7^4*t^2+80*7^2*t
      +8)+p1*(-48*7^6*t^3-65*7^5*t^2-24*7^4*t-32*7)-7^7*t^4+16*7^6
      *t^3+199*7^4*t^2+664*7^2*t+740):
  proof4_5_4 := provemodfuncGAMMA0UpETAidBATCH(EP4_5_4,7,RHS4_5_4,14);
```

```
*** There were NO errors.
*** o each EP in gcombo is an MF on Gamma[0](98)
*** o Each term in the etacombo is a  modular function on
      Gamma0(14).
*** o We also checked that the total order of
      each term etacombo was zero.
*** To prove the identity U[7](EP)=etacombo we need to show
    that v[oo](ID) > 14    This means checking up to q^(15).
We find that LHS - RHS is
```

$$O(q^{42})$$

$$proof4_5_4 := [1, -14, O(q^{42})] \quad \textbf{(32)}$$

```
> # GROUP 5, IDENTITY 5: U_B4(p1*t^(-5))
  EP4_5_5 := expand(B4*p1/t^5):
  RHS4_5_5 := expand(p0*(-8*7^9*t^5-64*7^8*t^4-80*7^7*t^3-216*7^5*t^2
      -80*7^3*t-8*7)+p1*(8*7^8*t^4+9*7^8*t^3+72*7^6*t^2+24*7^5*t
      +32*7^2)-7^9*t^5-8*7^8*t^4-31*7^7*t^3-248*7^5*t^2-664*7^3*t
      -740*7):
  proof4_5_5 := provemodfuncGAMMA0UpETAidBATCH(EP4_5_5,7,RHS4_5_5,14);
```

```
*** There were NO errors.
*** o each EP in gcombo is an MF on Gamma[0](98)
*** o Each term in the etacombo is a  modular function on
      Gamma0(14).
*** o We also checked that the total order of
      each term etacombo was zero.
*** To prove the identity U[7](EP)=etacombo we need to show
    that v[oo](ID) > 17    This means checking up to q^(18).
We find that LHS - RHS is
```

$$O(q^{45})$$

$$proof4_5_5 := [1, -17, O(q^{45})] \quad \textbf{(33)}$$

```
> # GROUP 5, IDENTITY 6: U_B4(p1*t^(-6))
  EP4_5_6 := expand(B4*p1/t^6):
```

```
RHS4_5_6 := expand(p0*(-8*7^11*t^6-8*7^10*t^5+40*7^9*t^4+400*7^7*t^3
    +1080*7^5*t^2+400*7^3*t+40*7)+p1*(8*7^10*t^5+7^10*t^4-40*7^8
    *t^3-360*7^6*t^2-120*7^5*t-160*7^2)-7^11*t^6-8*7^10*t^5-7^10*t^4
    +120*7^7*t^3+1240*7^5*t^2+3320*7^3*t+4720*7):
proof4_5_6 := provemodfuncGAMMA0UpETAidBATCH(EP4_5_6,7,RHS4_5_6,14);
```

```
*** There were NO errors.
*** o each EP in gcombo is an MF on Gamma[0](98)
*** o Each term in the etacombo is a  modular function on
      Gamma0(14).
*** o We also checked that the total order of
      each term etacombo was zero.
*** To prove the identity U[7](EP)=etacombo we need to show
    that v[oo](ID) > 20    This means checking up to q^(21).
We find that LHS - RHS is
```

$$O(q^{48})$$

$$proof4_5_6 := [1, -20, O(q^{48})] \quad \textbf{(34)}$$

```
> # GROUP 5, IDENTITY 7: U_B4(p1*t^(-7))
  EP4_5_7 := expand(B4*p1/t^7):
  RHS4_5_7 := expand(p0*(-8*7^13*t^7-8*7^12*t^6-136*7^9*t^4-1360*7^7*t^3
    -3672*7^5*t^2-1360*7^3*t-136*7)+p1*(8*7^12*t^6+6*7^11*t^5-19*7
    ^9*t^4+818*7^7*t^3+1167*7^6*t^2+2798*7^4*t+11*7^4+t^(-1))+6
    *7^13*t^7+20*7^12*t^6+272*7^10*t^5+1893*7^8*t^4+435*7^7*t^3
    -2953*7^5*t^2-10620*7^3*t-27029*7):
  proof4_5_7 := provemodfuncGAMMA0UpETAidBATCH(EP4_5_7,7,RHS4_5_7,14);
```

```
*** There were NO errors.
*** o each EP in gcombo is an MF on Gamma[0](98)
*** o Each term in the etacombo is a  modular function on
      Gamma0(14).
*** o We also checked that the total order of
      each term etacombo was zero.
*** To prove the identity U[7](EP)=etacombo we need to show
    that v[oo](ID) > 24    This means checking up to q^(25).
We find that LHS - RHS is
```

$$O(q^{52})$$

$$proof4_5_7 := [1, -24, O(q^{52})] \quad \textbf{(35)}$$

```
> # GROUP 6, IDENTITY 1: U_B4(1) = U_7(1) = 1, directly from the definition of U_7.

> # GROUP 6, IDENTITY 2: U_B4(t^(-1))
  EP4_6_2 := normal(B4/t):
```

```
RHS4_6_2 := -7*t-4:
proof4_6_2 := provemodfuncGAMMA0UpETAidBATCH(EP4_6_2, 7, RHS4_6_2, 14);
```

```
*** There were NO errors.
*** o each EP in gcombo is an MF on Gamma[0](98)
*** o Each term in the etacombo is a  modular function on
      Gamma0(14).
*** o We also checked that the total order of
      each term etacombo was zero.
*** To prove the identity U[7](EP)=etacombo we need to show
    that v[oo](ID) > 3    This means checking up to q^(4).
We find that LHS - RHS is
```

$$\mathrm{O}(q^{31})$$

$$proof4_6_2 := [1, -3, \mathrm{O}(q^{31})] \tag{36}$$

```
> # GROUP 6, IDENTITY 3: U_B4(t^(-2))
  EP4_6_3 := normal(B4/t^2):
  RHS4_6_3 := -7^3*t^2+20:
  proof4_6_3 := provemodfuncGAMMA0UpETAidBATCH(EP4_6_3, 7, RHS4_6_3, 14);
```

```
*** There were NO errors.
*** o each EP in gcombo is an MF on Gamma[0](98)
*** o Each term in the etacombo is a  modular function on
      Gamma0(14).
*** o We also checked that the total order of
      each term etacombo was zero.
*** To prove the identity U[7](EP)=etacombo we need to show
    that v[oo](ID) > 6    This means checking up to q^(7).
We find that LHS - RHS is
```

$$\mathrm{O}(q^{34})$$

$$proof4_6_3 := [1, -6, \mathrm{O}(q^{34})] \tag{37}$$

```
> # GROUP 6, IDENTITY 4: U_B4(t^(-3))
  EP4_6_4 := normal(B4/t^3):
  RHS4_6_4 := -7^5*t^3-88:
  proof4_6_4 := provemodfuncGAMMA0UpETAidBATCH(EP4_6_4, 7, RHS4_6_4, 14);
```

```
*** There were NO errors.
*** o each EP in gcombo is an MF on Gamma[0](98)
*** o Each term in the etacombo is a  modular function on
      Gamma0(14).
*** o We also checked that the total order of
      each term etacombo was zero.
```

```
*** To prove the identity U[7](EP)=etacombo we need to show
    that v[oo](ID) > 9    This means checking up to q^(10).
We find that LHS - RHS is
```

$$\mathrm{O}(q^{37})$$

$$proof4_6_4 := [1, -9, \mathrm{O}(q^{37})] \tag{38}$$

```
> # GROUP 6, IDENTITY 5: U_B4(t^(-4))
  EP4_6_5 := normal(B4/t^4) :
  RHS4_6_5 := -7^7*t^4 + 260 :
  proof4_6_5 := provemodfuncGAMMA0UpETAidBATCH(EP4_6_5, 7, RHS4_6_5, 14);
*** There were NO errors.
*** o each EP in gcombo is an MF on Gamma[0](98)
*** o Each term in the etacombo is a  modular function on
      Gamma0(14).
*** o We also checked that the total order of
      each term etacombo was zero.
*** To prove the identity U[7](EP)=etacombo we need to show
    that v[oo](ID) > 13    This means checking up to q^(14).
We find that LHS - RHS is
```

$$\mathrm{O}(q^{41})$$

$$proof4_6_5 := [1, -13, \mathrm{O}(q^{41})] \tag{39}$$

```
> # GROUP 6, IDENTITY 6: U_B4(t^(-5))
  EP4_6_6 := normal(B4/t^5) :
  RHS4_6_6 := -7^9*t^5 + 68*7 :
  proof4_6_6 := provemodfuncGAMMA0UpETAidBATCH(EP4_6_6, 7, RHS4_6_6, 14);
*** There were NO errors.
*** o each EP in gcombo is an MF on Gamma[0](98)
*** o Each term in the etacombo is a  modular function on
      Gamma0(14).
*** o We also checked that the total order of
      each term etacombo was zero.
*** To prove the identity U[7](EP)=etacombo we need to show
    that v[oo](ID) > 16    This means checking up to q^(17).
We find that LHS - RHS is
```

$$\mathrm{O}(q^{44})$$

$$proof4_6_6 := [1, -16, \mathrm{O}(q^{44})] \tag{40}$$

```
> # GROUP 6, IDENTITY 7: U_B4(t^(-6))
  EP4_6_7 := normal(B4/t^6) :
```

$RHS4_6_7 := -7\text{^}11 * t\text{^}6 - 2392 * 7 :$

$proof4_6_7 := provemodfuncGAMMA0UpETAidBATCH(EP4_6_7, 7, RHS4_6_7, 14);$

```
*** There were NO errors.
*** o each EP in gcombo is an MF on Gamma[0](98)
*** o Each term in the etacombo is a  modular function on
      Gamma0(14).
*** o We also checked that the total order of
      each term etacombo was zero.
*** To prove the identity U[7](EP)=etacombo we need to show
    that v[oo](ID) > 19    This means checking up to q^(20).
We find that LHS - RHS is
```

$$O(q^{47})$$

$$proof4_6_7 := [1, -19, O(q^{47})] \quad \textbf{(41)}$$

>